Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/04/2024
PI Name:	Laiakis, Evagelia Ph.D.		
Project Title:	Alterations in Energy Metabolism Pathways in Skeletal Muscle in Relation to Microgravity Analog and Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ecl28@georgetown.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	202-687-3114
Organization Name:	Georgetown University		
PI Address 1:	Department of Oncology		
PI Address 2:	3970 Reservoir Rd NW, NRB EP11		
PI Web Page:			
City:	Washington	**State:**	DC
Zip Code:	20007-2126	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 HERO 80JSC020N0001-FLAGSHIP, OMNIBUS1 Human Research Program: Crew Health Appendix A; Omnibus1-Appendix B
Start Date:	01/09/2023	**End Date:**	03/31/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/31/2026 per S. Mack-Phillips/JSC (Ed., 3/12/25). NOTE: End date changed to 03/31/2025 per NSSC information (Ed., 11/27/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Beheshti, Afshin Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	80NSSC23K0366		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Muscle atrophy is a well documented response to microgravity (μG) during short-term missions within low-Earth orbit (LEO). Recent studies have identified that low dose, charged-particle radiation characteristic of galactic cosmic rays (GCR) and high dose solar particle events (SPE) outside LEO damage skeletal muscles. This damage may deplete the energetic metabolic capacity of the tissue and impair astronaut performance. Thus, due to the combined challenges of μG and space radiation, countermeasures for muscle atrophy on long-duration missions outside LEO, other than daily exercise, need to be developed to preserve muscle mass and function. In collaboration with Jeffrey Willey, Ph.D., we analyzed muscle tissues (gastrocnemius, quadricep) from mice flown to the International Space Station, and identified significant perturbations in the tricarboxylic acid (TCA) cycle and fatty acid beta-oxidation intermediates, among others. Importantly, pantothenic acid (Vitamin B5) exhibited significantly decreased levels – affecting multiple energy related metabolic pathways, as it serves as the precursor for coenzyme CoA, an important mediator for mitochondrial related metabolism. Radiation exposure can also lead to perturbations in energy metabolism that could be additive or synergistic with μG. We hypothesize that the space environment with ground based analogs (simulated μG via hindlimb unloading/HU and simulated space radiation) will lead to exacerbated alterations in intramuscular energy metabolism in mice, such as in the TCA cycle, fatty acid beta-oxidation, and cofactors, among others.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work evaluated the extent of altered energy metabolism that contributes to muscle atrophy. Specifically, we investigated the combination of space radiation and hindlimb unloading as a ground-based surrogate for microgravity simulation, and used metabolomics and lipidomics to determine energetic perturbations that could be used for potential mitigation during long term space missions. These studies will provide considerable insight in metabolism associated with muscle atrophy, with particular emphasis in energy and mitochondrial metabolism.
Task Progress:	Progress is presented according to the specific aims. Aim 1 focused on identifying perturbations in gastrocnemius (gastroc) and quadriceps (quad) from mice with hindlimb unloading, space radiation, or a combination. - Carnitine and acetylcarnitine show considerable perturbations that could impact the carnitine transport chain and subsequent fatty acid b-oxidation. - Carnitine and acetylcarnitine show higher perturbations in quad vs. gastroc, both more pronounced in galactic cosmic rays + hindlimb unloading (GCR+HU). Aim 2 focused on determining the extent of perturbations, and whether those are seen to exert additive or synergistic effects. - Multiple reaction monitoring (MRM) methods were designed for 13 cofactors on a sensitive liquid chromatography mass spectrometry system (UPLC-Xevo TQS Waters) instrument. - Relative or absolute quantification will be conducted on all tissues diluted 1:100. - Assessment total protein levels with Western Blots of select enzymes, mostly mitochondrial related.
Bibliography Type:	Description: (Last Updated: 11/27/2024)
Conference Materials (Downloadable)	Kosowski E, Mak TD, Blaber E, Willey JS, Beheshti A, Laiakis EC. "Alterations in energy metabolism pathways in skeletal muscle in relation to microgravity." 70th Annual Meeting of the Radiation Research Society (RRS), Tucson, Arizona, September 15-18, 2024. , Sep-2024 RRS NASA 2024_Laiakis_v2.pdf